

The Performance Commons — List of Figures and Tables

Read top to bottom, the visuals carry the paper's argument on their own: magnitude → mechanism → the same failure everywhere → the same problem, distributed → proof → destination → already underway → the program. Each entry notes the beat it carries and whether it rests on measured data or is an argued/conceptual construct. (The defections and the diagnosis of the herd live in §2; Figure 2 carries §2.2's measured centerpiece.)

Table 1 — The efficiency gap, quantified. (§1) The measured cost of generality: ASIC 1×, CGRA ~2.8× area, FPGA ~20–35× area / ~10–14× power, GPU ~3–15× energy on regular kernels (TPU ~30× perf/W over its contemporary GPU), CPU ~100–500× energy. *Beat: the gap is real — most code sits far below its own architecture's frontier; these ratios are the separate physical floor.* — **quantitative, sourced** (Hameed; Kuon & Rose; Prabhakar).

Figure 1 — The general-purpose vs. custom trade-off is mostly an effort artifact. (§1.2) The effort gap, concentrated on reconfigurable substrates; close it and a CGRA sits within ~2.8× of an ASIC with full flexibility. *Beat: that gap is effort, not physics — the hardware question (Q1).* — **anchored endpoints**; finite-effort curve illustrative.

Figure 2 — The embargo natural experiment, drawn. (§2.2) Top-US-vs-top-Chinese Arena gap: 9.3% (Jan 2024) → 1.7% (Feb 2025) → 2.7% (2026), across export controls (Oct 2022, tightened Oct 2023) that held the training-hardware lead at ~4 years (Epoch AI). *Beat: chips were restricted; capability parity happened anyway — the herd's effort and variance, not silicon, was binding.* — **quantitative** (Stanford AI Index 2025/2026; Epoch AI); single-metric series, leaderboard limits noted.

Table 2 — One pattern, every layer of the stack. (§3) Foundry → CAD/EDA → IP → ISA → firmware → kernels → drivers → measurement data → distributed compute, each with its open counter-trend. *Beat: it is not one gap — the same coordination failure recurs from the foundry to the ISA.* — **qualitative + quantitative.**

Figure 3 — Distributed computing is a collective-action problem, not a hardware limit. (§4) The coordination gap by looseness of coupling; naive methods waste almost all of a loose federation's compute, while methods built for loose coupling (DiLoCo: ~500× less communication, ~90% utilization across continents) reclaim it. *Beat: the same failure one level up — the wasted-supply side of the software question (Q2).* — **anchored to DiLoCo / OpenDiLoCo**; the underlying waste (12–18% average server utilization, up to ~30% "comatose") is sourced (NRDC / Anthesis).

Figure 4 — Machine demand has no floor. (§4) The value of the marginal compute-hour against cumulative demand: human demand saturates, machine demand extends toward arbitrarily low value — past datacenter economics (~\$0.50/hr) into the range only cheap idle compute (~\$0.08/hr, ~6× cheaper) can serve. *Beat: why solving it is worth the effort — unbounded machine demand makes the otherwise-wasted resource valuable.* — **conceptual**; the cost anchors are from Omerta's reliability-market simulation, the task-value ladder from its economic analysis.

Table 3 — The entry cost of first silicon. (§5.3) Full mask set (>\$500k) → academic MPW block (€2,000–5,900) → chipIgnite (~\$10k) → Tiny Tapeout (\$150–300, packaged, open flow, solo) → the same with an LLM agent (eighteen high-schoolers, ninety minutes). *Beat: the commons has already collapsed the entry cost of hardware specialization by orders of magnitude.* — **quantitative, sourced** (AnySilicon; Europractice 2021 list; Tiny Tapeout's IEEE paper; Krupp, Venn & Wehn 2026); subsidies prop the lowest tiers.

Figure 5 — Participation through the platform's death. (§5.3) Designs per Tiny Tapeout shuttle, 2022 → mid-2026, colored by foundry, with the March 2025 Efabless shutdown marked; the next shuttle was the largest ever and the following year out-produced any prior one. *Beat: the fitness argument's natural experiment, measured — no single point of strategic death.* — **quantitative** (tinytapeout.com); demand per run is spiky and partly subsidy-driven.

Figure 6 — Where the optimum lands as the commons matures. (§8.3) A phase diagram over commons-maturity-over-time × workload-diversity; the reconfigurable-optimal region sweeps into territory that today defaults to general-purpose silicon. *Beat: over time the optimum for real workloads migrates into reconfigurable silicon — not fixed ASICs.* — **conceptual**; the boundary is exactly what a bigger whole-system characterization (Appendix A's experiment) would pin down.

Figure 7 — Reconfigurable hardware is already moving in; the commons is the inflection. (§8.3) A decade of reconfigurable entry into the datacenter — Catapult, Intel–Altera \$16.7B, AWS F1, AMD–Xilinx \$49B, the AI-accelerator wave — with a projected inflection as the programmability gap closes. *Beat: the migration is measurable and already expensive; machine intelligence is the accelerant.* — **real milestones + acquisition values**; share curve illustrative, post-2025 a projection.

Table A1 — The program: four efforts, two questions, honest state. (Appendix A) Omerta (pool idle compute), the project-manager coordination layer (work on the pool), coherence (the inoculation thesis), and the reconfigurable demonstrator (the Q1 experiment) —

latest progress and open items for each, stated plainly. *Beat: the questions are under active attack, and the breadth of the program is itself evidence of the §6 multiplier.* — **status report, self-reported**; states verified against the repos at the time of writing.

Table A2 — The wedge economics. (Appendix A) macOS CI and inference prices, mid-2026: GitHub macOS runner \$3.72/hr (10× Linux) and EC2 Mac \$0.65/hr (24-hr minimum) vs ~\$0.006/hr idle-Mac electricity; 70B unified-memory inference ~\$0.60/Mtoken vs \$0.40 cheapest API and ~\$0.06 batched-4090 small models. *Beat: the Apple wedge prices where licensing and unified memory are the moat.* — **quantitative** (vendor price lists, published benchmarks); spot rates fluctuate.

How the sequence maps to the prose

Visual	Section	Argument beat
Table 1	§1.1	the gap is real and large
Figure 1	§1.2	the gap is effort, not physics (Q1: hardware)
—	§2	privilege → the central claim: continuous optimization is an under-provided public good (prose only)
Figure 2	§2.2	the embargo natural experiment: chips restricted, parity anyway
Table 2	§3	the same failure is everywhere in the stack
Figure 3	§4	distributed computing is the same problem — the wasted supply (Q2: software)
Figure 4	§4	unbounded machine demand makes the idle resource valuable
Table 3	§5.3	the entry cost of first silicon has collapsed
Figure 5	§5.3	the commons survives its platform's death, measured
—	§6–§7	(AI multiplies the commons; the vendor is the aggregation point — prose only)
Figure 6	§8.3	where the optimum lands as the commons matures
Figure 7	§8.3	and it is already underway
—	§9	honest limits (prose only)
—	Conclusion	build collaboration-native hardware (prose only)
Table A1	Appendix A	the program in detail, honestly
Table A2	Appendix A	the wedge economics, priced

The "needs a bigger project" items (flagged for honesty)

- **Figure 6's boundary** — the real position of the ASIC / CPU / CGRA optimal regions needs the whole-system workload characterization §9 calls the prerequisite (achieved-vs-achievable parallelism, dataflow locality, phase and input statistics across the real consumer mix).
- **Figure 4's demand curve** — conceptual; the ~\$0.50 vs ~\$0.08 cost anchors are from Omerta's reliability-market simulation and the task-value ladder from its economic-analysis notes. Omerta itself is partially built: a substantial, well-tested mesh-networking layer (encryption, NAT traversal, gossip discovery, tunnelling, a mesh-native virtual network, plus Terraform-managed bootstrap/STUN servers) and working ephemeral-VM and simulation components, but the end-to-end compute loop is broken mid-migration and the trust/payment chain is validated in simulation rather than running live (two of six transaction types tested).
- **Figure 7's share curve and projection** — the adoption trend is illustrative; the dated milestones and the \$16.7B / \$49B acquisition values are real.
- **Figure 1's finite-effort curve** — illustrative; only the realizable-frontier endpoints (the intrinsic fabric tax) are measured.